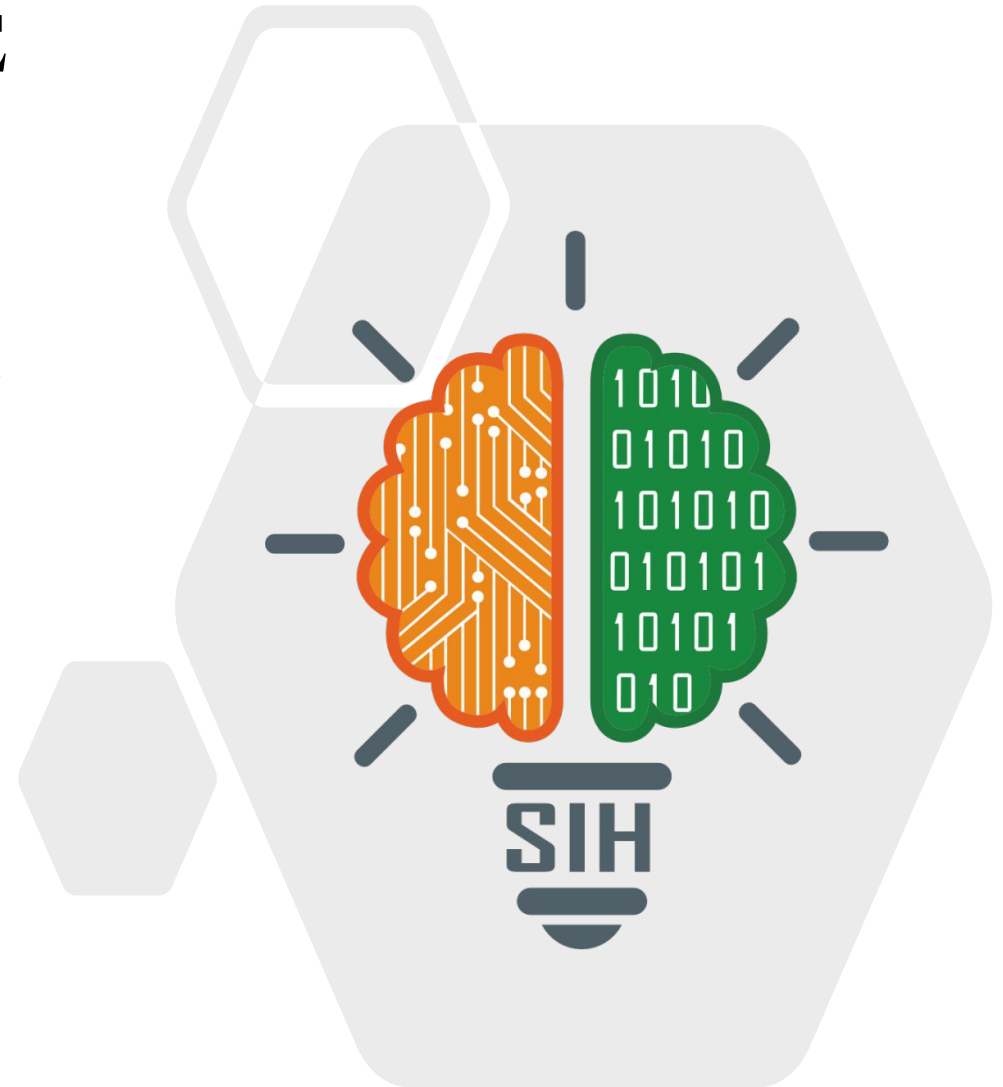


TITLE PAGE

- **Problem Statement ID** – SIH26123
- **Problem Statement Title**- Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 157578
- **Team Name**- FLEEX_NZD



Problem:

Centralized fleet coordination can become a single point of failure. When the central fleet service or its communication path becomes unavailable, AMRs may lose coordination, wait unnecessarily, leave tasks unresolved, or compete for constrained warehouse areas.

Proposed Solution:

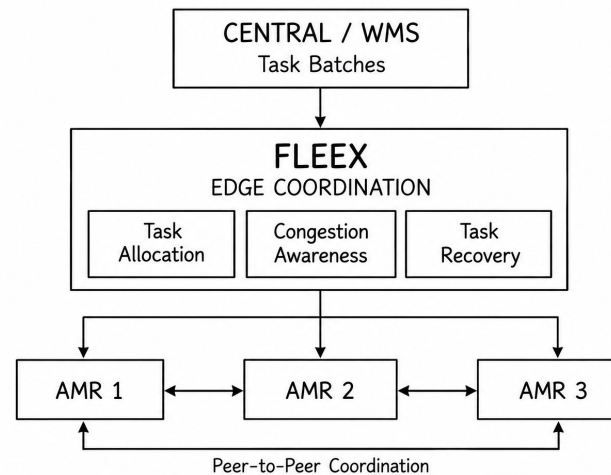
- The central system sends **batches of warehouse tasks** to the FLEEX edge. AMRs then exchange **fleet state and task information** through ROS 2 + Zenoh to coordinate task execution without requiring the central system for every decision.
- AMRs **distribute the available tasks and coordinate access to constrained zones**. If a robot fails or a task becomes blocked, **heartbeat/TTL mechanisms enable task recovery and reassignment** within the fleet.
- **Edge-AI uses LightGBM to estimate congestion-related task cost**, helping the fleet choose better task/route options, while **navigation and safety remain handled by**

Our Idea:

AMRs communicate peer-to-peer and make local coordination decisions for **task allocation, congestion, zone access and task recovery**, while the central system remains available for **interoperability, monitoring and higher-level task management**.

Innovation/Uniqueness:

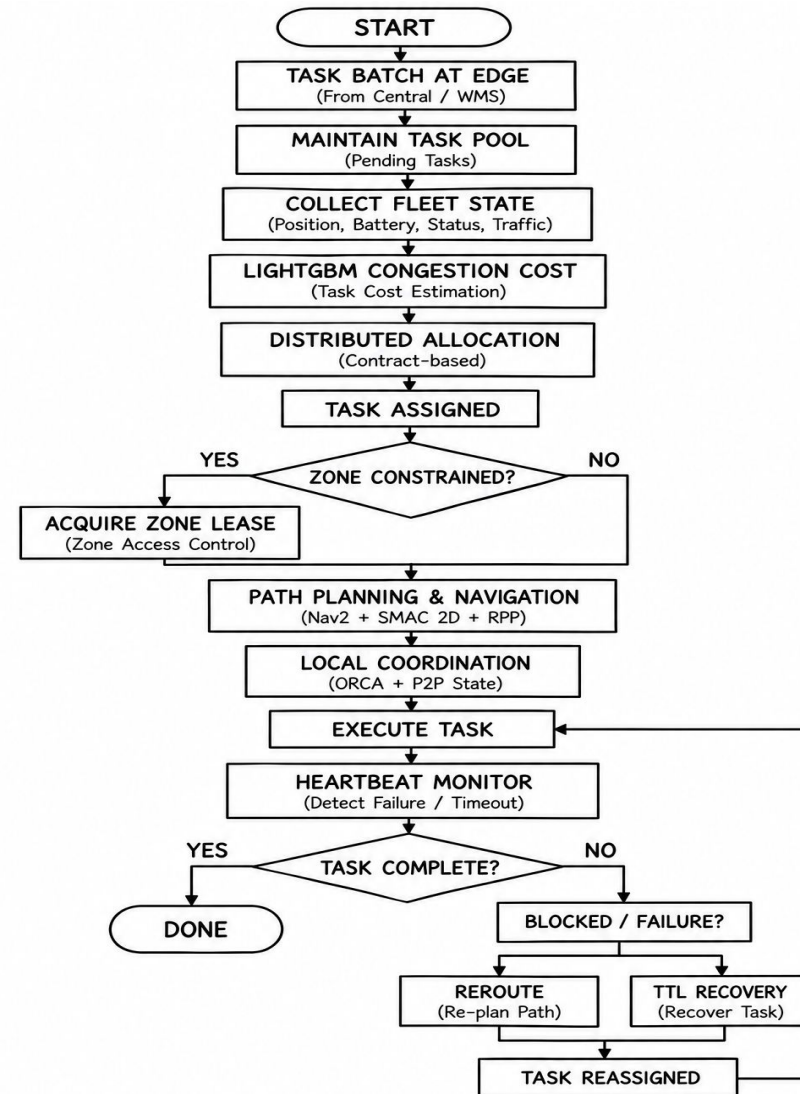
- **Distributed fleet coordination** reduces continuous dependence on centralized fleet decisions.
- **Fault-tolerant task recovery** uses heartbeat and time to live(TTL) to recover unfinished tasks after robot failures.
- **Congestion-aware Edge-AI** uses LightGBM to improve task allocation using local traffic conditions.
- **Local zone coordination** uses zone leases to manage access to narrow aisles and other constrained areas.



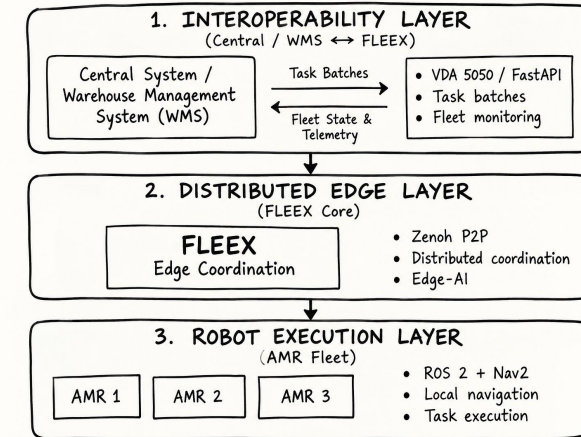
Software Stack & Design Choices

- **Robot Operating System 2 (ROS 2) Jazzy:** Modular robot software with integration for navigation, sensors and controllers.
- **Zenoh Peer-to-Peer (P2P):** Lightweight communication for distributed fleet coordination without continuous dependence on a central fleet controller.
- **Navigation 2 (Nav2) + SMAC 2D + Regulated Pure Pursuit (RPP):** Global path planning and local path following for autonomous AMR navigation.
- **Optimal Reciprocal Collision Avoidance (ORCA) + Zone Leases:** Local collision avoidance and controlled access to constrained warehouse zones.
- **Distributed Task Allocation + LightGBM:** Local task assignment using congestion-aware task-cost estimation.
- **Heartbeat + Time-to-Live (TTL) Recovery:** Detects robot failure and enables unfinished tasks to be recovered and reassigned.
- **Gazebo Harmonic:** Multi-AMR simulation

Flow Chart:



3 Layer Approach:



Video Demo: [Link](#)
Github Repo: [Link](#)

**Above 40% of
Prototype is Done.**

Feasibility

- **Simulation-First Readiness:** Validated end-to-end multi-AMR warehouse physics using **ROS 2 Jazzy + Gazebo Harmonic**.
- **Edge Compute Feasible:** Deployable on low-power, low-cost hardware (**Raspberry Pi 4 / NVIDIA Jetson Nano**).
- **Modular Integration:** Separates **Zenoh P2P** messaging, **Edge-AI** allocation, **Nav2**, and **ORCA** safety for easy maintenance.
- **Proven Edge-Case Recovery:** Successfully resolves real-time conflicts, deadlocks, chokepoints, and aisle blockages.

Operational Viability

- **Server-Independent Operation:** Safety-critical navigation runs locally on AMRs, ensuring continuous uptime during network outages.
- **Scalable P2P Networking:** Range- and zone-aware communication minimizes network traffic and prevents Wi-Fi congestion.
- **Clear Deployment Pipeline:** Seamless path from single-workstation simulation to distributed edge nodes on warehouse LAN.
- **Target Operational Metrics:** Achieves **0 collisions** and a **< 20% lower task completion time** vs. traditional stop-and-wait methods

Challenges & Mitigation

- **Compute Bottlenecks:** Lightweight, bounded Edge-AI keeps CPU loads minimal.
- **Wi-Fi Dropouts:** Range-aware P2P caching maintains local robot state updates.
- **Aisle Deadlocks:** Priority reservation and Nav2 dynamic rerouting clear blockages instantly.

Direct Impact on Target Users

- **Warehouse Operators:** Eliminates waiting delays via local task allocation and coordination.
- **Fleet Managers:** Automated recovery from deadlocks, chokepoints, and blocked routes.
- **AMR Fleets:** Continuous local operation during network outages or server drops.
- **Overall Operations:** Higher throughput and resilience with zero reliance on central safety servers.

Strategic Impact

- **Decentralized Control:** Removes single-point server dependency for safety-critical fleet operations.
- **High Fault Tolerance:** Ensures unbroken local coordination through network drops and server outages.
- **Edge Intelligence:** Shifts task allocation, navigation, and safety directly onto onboard robot compute.
- **Scalable Integration:** Seamless transition path: Simulation -> Edge LAN -> Physical Warehouse

Economic & Operational Benefits

- **Reduced Server Capex:** Cuts infrastructure costs via local peer-to-peer coordination.
- **Maximized Uptime:** Dynamic rerouting and reassignment instantly clear aisle blockages.
- **Low-Cost Scaling:** Lightweight stack seamlessly handles growing AMR fleet sizes.
- **100% Open-Source:** Uses **ROS 2, Nav2, Zenoh, and Gazebo** to eliminate licensing fees.

Gap & Problem Identification

- Centralized coordination creates [single-point failure](#) under communication loss.
- Warehouse AMRs require [conflict](#), [deadlock and blockage recovery](#).
- Need for [resilient local coordination](#).

TECHNOLOGY BENCHMARKING

- [Centralized](#): global view, infrastructure dependent
- [Distributed](#): local decisions, communication resilient
- FLEEX**: distributed robot safety + centralized monitoring

WAREHOUSE RESEARCH & VALIDATION

- Lifelong MAPF**: Continuous task planning and conflict resolution for warehouse fleets.
- Deadlock Avoidance**: Recent AGV research uses **historical path data** for warehouse deadlock prevention.
- FLEEX Validation**: Targets **zero collisions** and **≥20% lower task completion time** vs. stop-and-wait.

LITERATURE SURVEY

- [MRTA](#): decentralized auctions & consensus.
- [ORCA](#): local multi-robot collision avoidance.
- [MAPF](#): coordinated planning in shared spaces.
- [Lifelong MAPF](#): continuous warehouse task planning.

FLEEX RESEARCH GAP

Existing: Allocation + Planning + Avoidance + Deadlock Handling

FLEEX: P2P → Allocation → Local Safety → Recovery

RESEARCH REFERENCES

01. Jia & Meng (2013) | [MRTA](#) Survey
02. Choi et al. (2009) | [Decentralized](#) Auctions
03. van den Berg et al. (2011) | [Collision](#) Avoidance
04. Li et al. (2021) | [Lifelong](#) MAPF
05. Zhang et al. (2025) | AGV [Deadlock Avoidance](#)